

MLFCS

Matrices: Definition, Examples & Matrix Multiplication

Rajesh Chitnis

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Lecture attendance code:

80113131

(start recording)



Today's plan

- ▶ Definition of a matrix
- ▶ Vector space of all $(m \times n)$ matrices with entries from $\mathbb{Q}$
 - ▶ Need to define addition of two matrices
 - ▶ Need to define multiplication of a matrix by a scalar
- ▶ Definition of matrix multiplication
- ▶ Identity for square matrices
- ▶ Computing inverse of a matrix



What is a matrix?

For us, matrix is a two-dimensional array whose entries come from a field (say $\mathbb{Q}$ or $\mathbb{R}$)

- ▶ We can read it row-wise
- ▶ or column-wise

A handwritten diagram of a 2x3 matrix. The matrix is enclosed in large blue parentheses and contains the numbers 1, 4, 9 in the first row and 3, 5, 12 in the second row. To the left of the matrix, a red arrow points down from 'Row 1' to 'Row 2'. Below the matrix, the labels 'c1', 'c2', and 'c3' are written in red. To the right of the matrix, a pink arrow points to the text 'This a (2x3) matrix'.

Size of a matrix

- ▶ Number of rows $\rightarrow 2$
 - ▶ Number of columns $\rightarrow 3$
- } # of entries = $2 \times 3 = 6$

How to denote entries of a matrix?

Handwritten notation for a 2x3 matrix. It shows $A_{2 \times 3} = (a_{ij})_{\substack{1 \leq i \leq 2 \\ 1 \leq j \leq 3}} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix}$. The matrix is written in blue ink.



Addition of two matrices over the field $\mathbb{Q}$

We can add two matrices if they have the same size

- ▶ Number of rows must be same
- ▶ Number of columns must also be the same

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$$

Matrix addition is not defined

This is done by adding the corresponding entries from each matrix!

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} 1 & 11 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 2 & 13 \\ 6 & 8 \end{pmatrix}$$

Example:



Multiplying a matrix by a scalar over the field $\mathbb{Q}$

Let A be any $(m \times n)$ matrix with entries from $\mathbb{Q}$

For any $r \in \mathbb{Q}$, we define the product of the scalar $r \in \mathbb{Q}$ and the matrix A as the $(m \times n)$ matrix $B := rA$ obtained as follows:

- ▶ Each entry of B is obtained by multiplying the corresponding entry of A by r
- ▶ That is, for each $1 \leq i \leq m$; $1 \leq j \leq n$ we define $b_{i,j} = r \times a_{i,j}$

$$3 \begin{pmatrix} 1 & 11 & 0 \\ 2 & 3 & 5 \end{pmatrix} = \begin{pmatrix} 3 & 33 & 0 \\ 6 & 9 & 15 \end{pmatrix}$$

Example:



Vector space of matrices over the field $\mathbb{Q}$

Another example of a vector space:

- ▶ Fix any $m, n \geq 1$

$$m=2 \text{ \& } n=3$$

$$\begin{pmatrix} \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet \end{pmatrix}$$

- ▶ Then the set of all $(m \times n)$ matrices whose entries are from $\mathbb{Q}$
 - ▶ Each vector in this vector space is a $(m \times n)$ matrix
 - ▶ Each scalar in this vector space is a rational number

To show this is a vector space, we need to verify the following 8 conditions

For any vectors $\vec{u}, \vec{v}, \vec{w} \in V$ and any scalars $r, s \in F$

- (1) Commutativity of vector addition: $\vec{u} \oplus \vec{v} = \vec{v} \oplus \vec{u}$
- (2) Associativity of vector addition: $(\vec{u} \oplus \vec{v}) \oplus \vec{w} = \vec{u} \oplus (\vec{v} \oplus \vec{w})$
- (3) Existence of Additive identity: $\vec{0} \oplus \vec{v} = \vec{v}$ $\leftarrow$ Matrix with all zero entries
- (4) Existence of additive inverse: for each $\vec{x}$, there exists $\vec{-x}$ such that $\vec{x} \oplus \vec{-x} = \vec{0}$
- (5) Associativity of multiplication of scalar & vector: $r(s\vec{v}) = (rs)\vec{v}$ $\uparrow$??
- (6) Distributivity of scalar sums: $(r + s)\vec{v} = r\vec{v} \oplus s\vec{v}$
- (7) Distributivity of vector sums: $r(\vec{u} \oplus \vec{v}) = r\vec{u} \oplus r\vec{v}$
- (8) Existence of identity of multiplication of scalar & vector: $1\vec{v} = \vec{v}$

First, we need to define two operations for above 8 conditions to make sense:

- ▶ Vector addition: for each $\vec{u}, \vec{v}$ a vector from V is assigned to $\vec{u} \oplus \vec{v}$ $\leftarrow$ slide 5
- ▶ Multiplication of a scalar by a vector: for each $s \in F$ and $\vec{v} \in V$, a vector from V is assigned to $s\vec{v}$ $\leftarrow$ slide 6



Definition of matrix multiplication

We now define matrix multiplication:

- ▶ Let A be an $m \times n$ matrix whose entry in row i and column j is given by a_{ij}
- ▶ Let B be an $n \times p$ matrix whose entry in row j and column k is given by b_{jk}
- ▶ Then the result of multiplying A and B is a ($m \times p$) matrix C whose entry c_{ik} in row i and column k is given by

$$c_{ik} = \sum_{j=1}^n a_{ij} \times b_{jk} = (a_{i1} \times b_{1k}) + (a_{i2} \times b_{2k}) + (a_{i3} \times b_{3k}) + \dots + (a_{in} \times b_{nk})$$

- ▶ That is, the entry in row i and column k of the matrix AB is defined to be the inner product of row i of A with column k of B

$$A = \begin{pmatrix} 1 & 3 & 7 \\ 2 & 4 & 9 \end{pmatrix}$$

2×3

$$B = \begin{pmatrix} 6 & 3 \\ 7 & 2 \\ 8 & 1 \end{pmatrix}$$

3×2

$$\text{let } C = AB = \begin{pmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{pmatrix} = \begin{pmatrix} (1 \times 6) + (3 \times 7) + (7 \times 8) & (1 \times 3) + (3 \times 2) + (7 \times 1) \\ (2 \times 6) + (4 \times 7) + (9 \times 8) & (2 \times 3) + (4 \times 2) + (9 \times 1) \end{pmatrix}$$



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- ▶ That is, the entry in row i and column k of the matrix AB is defined to be the inner product of row i of A with column k of B

Handwritten example:

$$A = \begin{pmatrix} 1 & 3 \\ 5 & 6 \end{pmatrix} \quad B = \begin{pmatrix} 0 & 3 \\ 7 & 11 \end{pmatrix}$$

Dimensions: A is 2×2 , B is 2×2 .

Find $AB = \begin{pmatrix} \text{Gray} \times \text{Blue} & \text{Gray} \times \text{Green} \\ \text{Red} \times \text{Blue} & \text{Red} \times \text{Green} \end{pmatrix} = \begin{pmatrix} 21 & 36 \\ 42 & 81 \end{pmatrix}$

Matrix multiplication is not commutative!

Definition of matrix multiplication:

- ▶ Let A be an $m \times n$ matrix whose entry in row i and column j is given by a_{ij}
- ▶ Let B be an $n \times p$ matrix whose entry in row j and column k is given by b_{jk}
- ▶ Then the result of multiplying A and B is a $(m \times p)$ matrix C whose entry c_{ik} in row i and column k is given by

$$c_{ik} = \sum_{j=1}^n a_{ij} \times b_{jk} = (a_{i1} \times b_{1k}) + (a_{i2} \times b_{2k}) + (a_{i3} \times b_{3k}) + \dots + (a_{in} \times b_{nk})$$

- ▶ That is, the entry in row i and column k of the matrix AB is defined to be the inner product of row i of A with column k of B

- ▶ For two matrices A and B , it is possible to have $AB \neq BA$

$$\left\{ \begin{array}{l} A_{2 \times 3} \quad B_{3 \times 4} \end{array} \right\} \rightarrow (AB)_{2 \times 4}$$

BA is not even defined

$$A = \begin{pmatrix} 1 & 3 \\ 5 & 6 \end{pmatrix} \quad B = \begin{pmatrix} 0 & 3 \\ 7 & 11 \end{pmatrix} \quad AB = \begin{pmatrix} 21 & 36 \\ 42 & 81 \end{pmatrix} \quad BA = \begin{pmatrix} 25 & 39 \\ 38 & 66 \end{pmatrix}$$



But matrix multiplication is associative!

- ▶ Let A be a $(m \times n)$ matrix
- ▶ Let B be a $(n \times p)$ matrix
- ▶ Let C be a $(p \times s)$ matrix
- ▶ Then $A(BC) = (AB)C$ ——— $(\star)$

Sanity checks:

- ▶ BC is well-defined and a $(n \times s)$ matrix
 - ▶ So $A(BC)$ is well-defined and a $(m \times s)$ matrix
 - ▶ AB is well-defined and a $(m \times p)$ matrix
 - ▶ So $(AB)C$ is well-defined and a $(m \times s)$ matrix
- 

▶ Proof is not very hard, but we will not cover it in this module!



Identity matrix for (2×2) matrices

Let $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ — $(\star)$

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

Let A be any (2×2) matrix over the field of rational numbers

► Show that $AI = A = IA$

$$AI = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} = A$$

$$IA = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} = A$$



Identity matrix for (3×3) matrices

$$\text{Let } I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$A = \begin{pmatrix} a & b & c \\ e & f & g \\ h & i & j \end{pmatrix}$$

Let A be any (3×3) matrix over the field of rational numbers



Show that $AI = A = IA$

→ Exercise



Inverse of a (2×2) matrix

Let A be a (2×2) matrix.

$$\rightarrow \begin{pmatrix} 1 & 2 \\ 3 & 8 \end{pmatrix}$$

Then a matrix B is said to be **inverse** of A if $BA = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$

How can we find inverse of a specific (2×2) matrix, say

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 8 \end{pmatrix} ? \rightarrow \text{Suppose } B = \begin{pmatrix} r & s \\ x & y \end{pmatrix} \text{ such that } BA = I$$

Expand left side

$$\begin{pmatrix} r & s \\ x & y \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 \\ 3 & 8 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \text{--- } (\star)$$

$$\begin{cases} r+3s=1 \\ 2r+8s=0 \end{cases}$$

$$\begin{cases} r+3y=0 \\ 2r+8y=1 \end{cases}$$

$$\begin{pmatrix} r+3s & 2r+8s \\ x+3y & 2x+8y \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Two equations in r & s

Two equations in x & y



Summary of the lecture

- ▶ Definition of a matrix
- ▶ Vector space of all $(m \times n)$ matrices with entries from $\mathbb{Q}$
 - ▶ Need to define addition of two matrices
 - ▶ Need to define multiplication of a matrix by a scalar
- ▶ Definition of matrix multiplication
 - ▶ Not always well-defined!
 - ▶ Is not commutative in general
 - ▶ Is always associative
- ▶ Identity for square matrices
- ▶ Computing inverse of a matrix
 - ▶ Can be computed using Gaussian elimination

